Manufacturing Work Order

48-inch Air-Cushioned Swing Check Valve • One-off ETO build

SYNTHETIC DEMONSTRATION — intended for Argus AI inspection; not an OEM-approved manufacturing or pressure-test instruction.

1 Order identification

FIELD	VALUE	FIELD	VALUE
Work order	WO-ARGUS-CVS48-001	Revision / issue	0.1 / 02 Nov 2026
Customer / project	Demo Water Transfer Station (fictional)	Quantity	1 complete valve
Equipment family	APCO CVS-6000D swing check valve	Nominal size	48 in (manufacturer size code 1200)
CAD reference	cvs-6000d-48-f1-ac(1).stp	Configuration	F1 flanged / AC air cushion (verify BOM)
Assembly identifier	CVS48-AC-DEMO-002	Order priority	Demonstration
Approx. handling class	~5-tonne class; verify certified lifting weight	Planned coating	Clean industrial blue, demo finish

2 Scope of supply

One assembled check-valve body with two flanged ends, bonnet and fasteners, check-disc/shaft internals as represented by the approved design, external lever/weight and air-cushion assembly, identification plate, finished coating, inspection records and handling protection. Exact material grades, pressure rating, drilling, gaskets, bolt grades, coating system and test values are to be confirmed against the approved purchase specification and manufacturer documentation.

3 Engineering document register

REF	DOCUMENT	USE / CONTROL
DWG-001	cvs-6000d-48-f1-ac(1).stp	Master spatial reference for CAD comparison; no guaranteed tolerances embedded
OEM-001	DeZURIK APCO CVS-6000 product/bulletin	Product-family verification; check selected configuration against PO/BOM
OEM-002	DeZURIK APCO CVS-6000 installation/maintenance manual D12059	OEM handling, assembly, operation and service instructions
INS-001	This work order — section 6	Demonstration acceptance rules and inspection IDs
TEST-001	Approved hydrostatic and functional test procedure — pending	Hold point; cannot mark as passed without actual records

4 Build objectives / acceptance boundary

Deliver a new, clean and correctly identified assembly whose external geometry, component presence, flange faces, lever linkage and visible fasteners conform to the approved engineering baseline. All findings must be supported by received evidence. A visually clean part is not proof of pressure integrity, internal dimensions or functional performance.

Traveler & inspection checkpoints

Inspection rule numbers below are DEMO work-order identifiers. They are not clause numbers from ASME, AWWA or the OEM manual.

5 Manufacturing traveler

STEP	OPERATION	EVIDENCE / HOLD POINT
10	Receive parts; verify PO, item identity, damage and required materials	Receiving checklist + ID photo
20	Check body, flanges and bonnet mating surfaces; clean and prepare	Pre-assembly photos; suspected indications logged
30	Assemble shaft/disc and bonnet; fasten to approved build instructions	Assembly record + bonnet fastener coverage photos
40	Install lever/weight and air-cushion linkage; verify attachment and clearance	Side-camera views + approved configuration comparison
50	Prepare and coat external surfaces with approved system	Paint inspection photos; coating certificate if available
60	Mount and verify nameplate/tag against work order	Readable close-up of installed nameplate
70	Perform dimensional/geometry survey and exterior inspection	Point-cloud registration + camera evidence
80	Perform OEM-approved seat/shell and movement tests	Recorded results; separate test procedure required
90	Review dossier, disposition NCRs and request release	QC authorization; do not auto-release product

6 Argus inspection rules — demo control points

RULE ID	CHECK / ACCEPTANCE CRITERION	EVIDENCE	SEVERITY
WO-R01	Both body flanges and bolt-hole patterns match the approved CAD orientation and are free of obvious geometric damage. Do NOT treat empty pipe-connection bolt holes as missing assembly bolts.	CAD + scan + camera	FAIL if verified
WO-R02	Machined sealing faces show no suspected crack-like linear indication, deep scoring or damaged seat land. A suspected hairline is an anomaly requiring QC/NDT confirmation; a photo alone is not proof of a crack.	High-resolution image / video	WARNING / QC review
WO-R03	All bonnet fasteners required by the approved BOM/drawing are present and seated; do not infer torque from images.	Bonnet images + CAD/BOM	FAIL if confirmed missing
WO-R04	External lever, pivot and air-cushion linkage have correct components and geometry; any suspected offset is flagged for measurement against approved tolerances.	Side images + scan + CAD	WARNING until measured
WO-R05	Painted exterior appears newly manufactured: no visible corrosion, runs, exposed substrate or unacceptable scuffs. Minor cosmetic deviations require review per finish spec.	Photos / video	WARNING / FAIL per spec
WO-R06	Nameplate exists, is attached in the designated position and visibly matches WO-ARGUS-CVS48-001 / CVS48-AC-DEMO-002, size 48 in and configuration where legible.	Nameplate photo + OCR	FAIL if verified mismatch
WO-R07	Required hydrostatic, seat	Test record	NOT EVALUATED until supplied

	leakage and cushion/movement tests have complete approved records within applicable limits. Without actual test data result is NOT EVALUATED.		

Inspection release & evidence

7 Required inspection media for Argus

EVIDENCE ID	SOURCE	REQUIRED CAPTURE / EXPECTED USE
EV-CAM-01	Video stream	Wide approach, both flanges, bonnet, complete linkage and final 360-degree coverage
EV-IMG-01	Still photo	Left/near sealing face, grazing illumination to reveal subtle linear indications
EV-IMG-02	Still photo	Bonnet ring and fasteners; sufficient resolution to count required bolts
EV-IMG-03	Still photo	External lever/air-cylinder linkage and fixture from side
EV-IMG-04	Still photo	Nameplate readable with assembly/WO reference
EV-PC-01	Laser point-cloud stream	Cumulative points aligned to STEP model with alignment/residual visualization
EV-TEST-01	Test records	Separate hydrostatic/seat/functional measurements, not generated by camera/scan

8 Review and hold points

GATE	ACCEPTANCE / DISPOSITION
HP-1 Identity	Correct nominal size, configuration and nameplate, or documented deviation
HP-2 External geometry	Flange, bonnet and linkage check outcomes recorded with traceable evidence
HP-3 Surface condition	Suspect flange indication routed to QC/NDT, not automatically treated as confirmed failure
HP-4 Test release	Actual test record attached and interpreted against approved limits
HP-5 Final disposition	Quality engineer signs final dossier; FAIL/WARNING/NOT EVALUATED never converted automatically to PASS

9 Demonstration scenario / seeded deviations

For the blue valve simulation, the following are optional planted conditions used to evaluate detection, not claims about a real manufactured unit: (a) a very subtle fixed-position hairline-like indication near one flange sealing face; (b) a slight linkage offset, if deliberately introduced and consistently reflected in media; (c) a bonnet fastener discrepancy, only if actually introduced. The simulator sends evidence only; it must never send ground-truth labels to the Argus inspection API.

10 Completion and signatures

RECORD	STATUS / SIGNATURE
Assembly completed / date	Pending _____
Dimensional inspection	Pending _____
Surface / NDT disposition	Pending _____
Pressure and functional test review	Pending _____
Final QC / customer release	ON HOLD — demonstration document only

Reference sources: DeZURIK APCO CVS-6000 product page and Bulletin 6000 (Oct 2025); DeZURIK manual D12059. These sources identify the family and configuration but do not by themselves approve demonstration-only tolerances, material certificates, or test procedures.